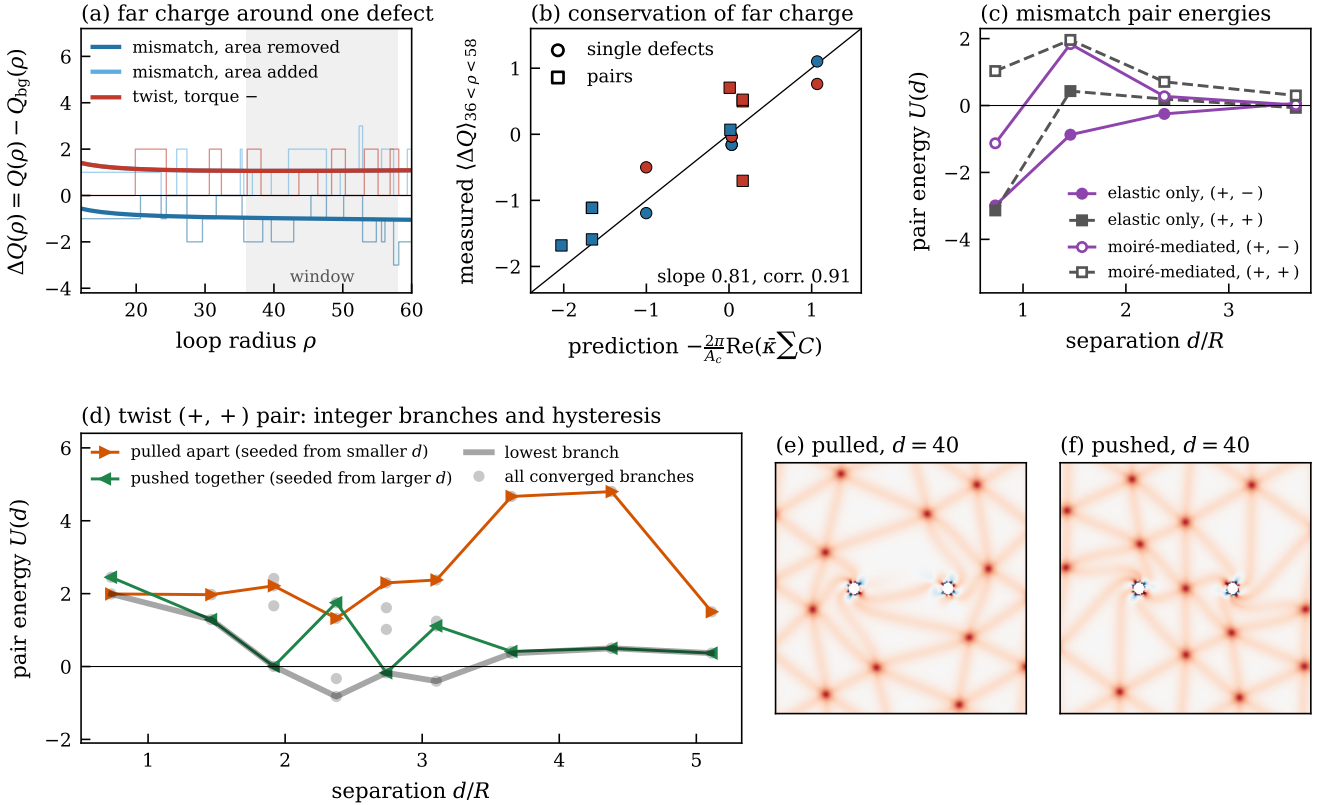

(a) far charge around one defect
mismatch, area removed
mismatch, area added
twist, torque −
window
$\Delta Q(\rho) = Q(\rho) - Q_{\mathrm{bg}}(\rho)$
loop radius ρ
(b) conservation of far charge
single defects
pairs
slope 0.81, corr. 0.91
measured $\langle \Delta Q \rangle_{36<\rho<58}$
prediction $-\frac{2\pi}{A_c}\mathrm{Re}(\bar{\kappa}\sum C)$
(c) mismatch pair energies
elastic only, (+, −)
elastic only, (+, +)
moiré-mediated, (+, −)
moiré-mediated, (+, +)
pair energy $U(d)$
separation d/R
(d) twist (+, +) pair: integer branches and hysteresis
pulled apart (seeded from smaller d)
pushed together (seeded from larger d)
lowest branch
all converged branches
pair energy $U(d)$
separation d/R
(e) pulled, $d = 40$
(f) pushed, $d = 40$